

Coding & Solution

5. Project Development Phase | Credit Card Approval Prediction

Prepared For	Prepared By	Domain	Date
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Development Workflow

The development phase implements the data science pipeline and connects the selected model to a Flask web application.

Training Pipeline Example

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report
from sklearn.ensemble import RandomForestClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.linear_model import LogisticRegression
from xgboost import XGBClassifier
import joblib

app = pd.read_csv("application_record.csv")
credit = pd.read_csv("credit_record.csv")

# Clean, merge, engineer labels, encode categories, and split data.
# X_train, X_test, y_train, y_test = ...

models = {
    "Logistic Regression": LogisticRegression(max_iter=1000),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "Random Forest": RandomForestClassifier(random_state=42),
    "XGBoost": XGBClassifier(eval_metric="logloss", random_state=42),
}

best_name, best_model, best_score = None, None, 0
for name, model in models.items():
    model.fit(X_train, y_train)
    preds = model.predict(X_test)
    score = accuracy_score(y_test, preds)
    if score > best_score:
        best_name, best_model, best_score = name, model, score

joblib.dump(best_model, "models/credit_card_model.pkl")
```

Flask Integration Example

```
from flask import Flask, render_template, request
import joblib
import pandas as pd

app = Flask(__name__)
model = joblib.load("models/credit_card_model.pkl")

@app.route("/")
def home():
    return render_template("index.html")

@app.route("/predict", methods=["POST"])
def predict():
    form_data = request.form.to_dict()
    input_df = pd.DataFrame([form_data])
    # Apply the same preprocessing used during model training.
```

```
prediction = model.predict(input_df)[0]
result = "Approved" if prediction == 1 else "Rejected"
return render_template("result.html", prediction=result)

if __name__ == "__main__":
    app.run(debug=True)
```